

Problem

Which of the following equations is called the state equation?

- (a) $\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bz}$
- (b) $\mathbf{y} = \mathbf{Cx} + \mathbf{Dz}$
- (c) $\mathbf{H}(s) = \mathbf{Y}(s)/\mathbf{Z}(s)$
- (d) $\mathbf{H}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$

Step-by-step solution

Step 1 of 1

The state variable model is used to analyse the complex systems with having different inputs and outputs and this model is also used in control theory and circuits.

The state of the system is the smallest set of quantities called the state variables. Write variable form of the state equation.

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bz}$$

Where,

$\dot{\mathbf{x}}(t)$ represents n state vectors.

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \vdots \\ \dot{x}_n(t) \end{bmatrix}$$

$\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ and $n \times m$ matrices respectively.

Dot represents the first derivative of with respect to time.

Thus, the correct option is .